

Checkpoint Answer Sheet

Name: _____ Date: _____

Complete independently. Use new data. Show all reasoning. Submit this PDF in **U2 L7 Checkpoint Submission**.

1. Spectrum relationships

A radiation sample has a wavelength of 620 nm. Convert the wavelength to meters and calculate its frequency.

Conversion	Substitution into $c = \text{wavelength} \times \text{frequency}$	Frequency with unit

2. Energy

Calculate the photon energy using $E = h \times \text{frequency}$ and your frequency from Question 1.

Substitution	Energy with unit

3. Compare

A second photon has a shorter wavelength. State how its frequency and energy compare with the 620 nm photon, and explain why.

Frequency comparison	Energy comparison	Reasoning

4. Interpret an atomic fingerprint

Reference spectrum lines occur at 410 nm, 434 nm, 486 nm, and 656 nm. An unknown sample shows lines at 410 nm, 486 nm, and 656 nm.

Claim: match or not a complete match	Evidence from the line sets	Reasoning about element identification

5. Application and limitation

Explain one real-world use of electromagnetic radiation or spectroscopy. Then identify one limitation or safety consideration that must be addressed.

Application	How the science is used	Limitation or safety consideration

Submission check: Your name is present; calculations show units; comparisons explain the relationship; spectrum claims cite exact line evidence.